

CHECKPOINT 3

Formulaire réponses

Exercice 1

Partie 1

Q.1.1.1

Copie(s) d'écran montrant que Lionel Lemarchand a les mêmes attributs de société que Kelly Rhameur.

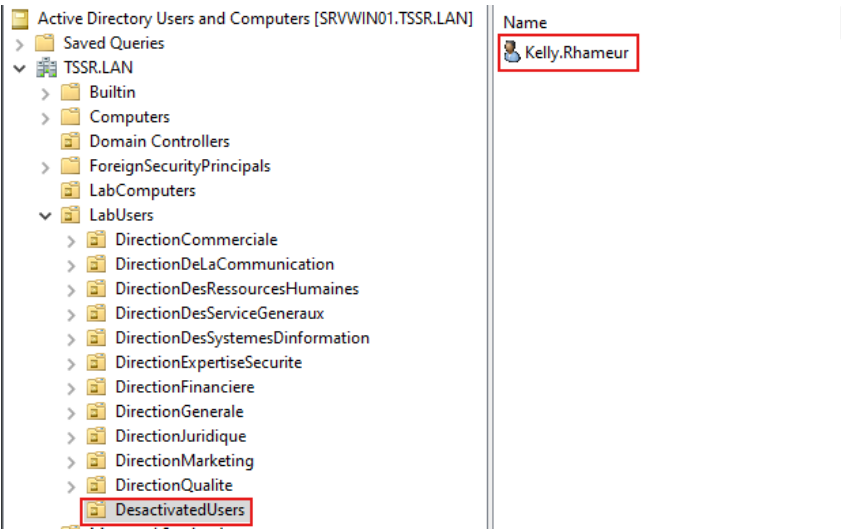
The image shows two side-by-side screenshots of the 'Properties' dialog box for users in Active Directory. The left window is for 'Lionel Lemarchand' and the right is for 'Kelly Rhameur'. Both windows have the 'General' tab selected. The 'Company' field in both is 'CyberOps'. The 'Department' field in both is 'Direction des Ressources Humaines'. The 'Job Title' field in both is 'Directeur des Ressources Humaines'. The 'Manager' field in both is 'Camille Martin'. The 'Direct reports' field in both is empty. The 'Web page' field in both is empty. The 'Telephone number' field in both is empty. The 'E-mail' field in both is 'Lionel.Lemarchand@TSSE.LAN' and 'Kelly.Rhameur@TSSR.LAN' respectively. The 'Display name' field in both is 'Lionel Lemarchand' and 'Kelly Rhameur' respectively. The 'Last name' field in both is 'Lemarchand' and 'Rhameur' respectively. The 'First name' field in both is 'Lionel' and 'Kelly' respectively. The 'Initials' field in both is empty. The 'Description' field in both is empty. The 'Office' field in both is empty. The 'Member Of' field in both is empty. The 'Dial-in' field in both is empty. The 'Environment' field in both is empty. The 'Sessions' field in both is empty. The 'Remote control' field in both is empty. The 'Remote Desktop Services Profile' field in both is empty. The 'COM+' field in both is empty. The 'General' field in both is empty. The 'Address' field in both is empty. The 'Account' field in both is empty. The 'Profile' field in both is empty. The 'Telephones' field in both is empty. The 'Organization' field in both is empty. The 'OK' button in both is highlighted.

Copie(s) d'écran montrant le changement coté management.

The image shows two side-by-side screenshots of the 'Properties' dialog box for users in Active Directory. The left window is for 'Lionel Lemarchand' and the right is for 'Kelly Rhameur'. Both windows have the 'General' tab selected. The 'Company' field in both is 'CyberOps'. The 'Department' field in both is 'Direction des Ressources Humaines'. The 'Job Title' field in both is 'Directeur des Ressources Humaines'. The 'Manager' field in both is 'Camille Martin'. The 'Direct reports' field in both is empty. The 'Web page' field in both is empty. The 'Telephone number' field in both is empty. The 'E-mail' field in both is 'Lionel.Lemarchand@TSSE.LAN' and 'Kelly.Rhameur@TSSR.LAN' respectively. The 'Display name' field in both is 'Lionel Lemarchand' and 'Kelly Rhameur' respectively. The 'Last name' field in both is 'Lemarchand' and 'Rhameur' respectively. The 'First name' field in both is 'Lionel' and 'Kelly' respectively. The 'Initials' field in both is empty. The 'Description' field in both is empty. The 'Office' field in both is empty. The 'Member Of' field in both is empty. The 'Dial-in' field in both is empty. The 'Environment' field in both is empty. The 'Sessions' field in both is empty. The 'Remote control' field in both is empty. The 'Remote Desktop Services Profile' field in both is empty. The 'COM+' field in both is empty. The 'General' field in both is empty. The 'Address' field in both is empty. The 'Account' field in both is empty. The 'Profile' field in both is empty. The 'Telephones' field in both is empty. The 'Organization' field in both is empty. The 'OK' button in both is highlighted.

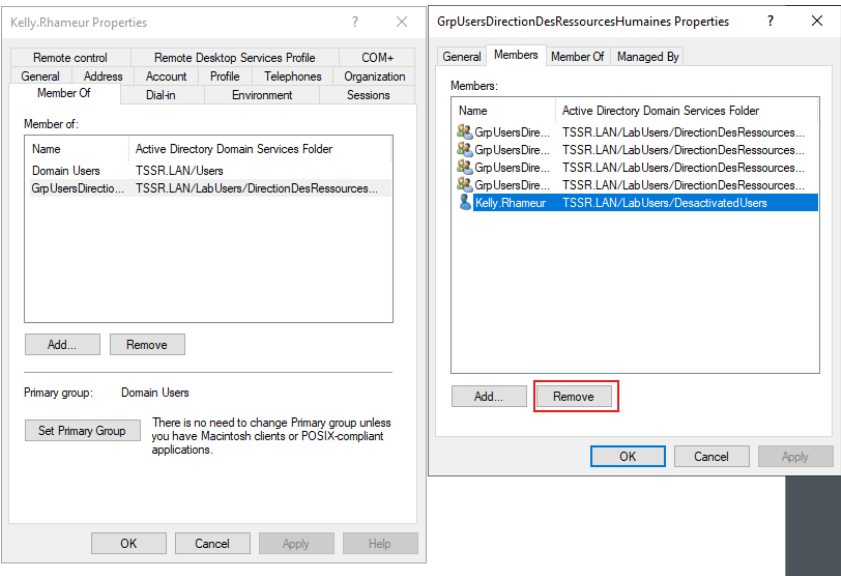
Q.1.1.2

Copie d'écran de l'OU DeactivatedUsers.



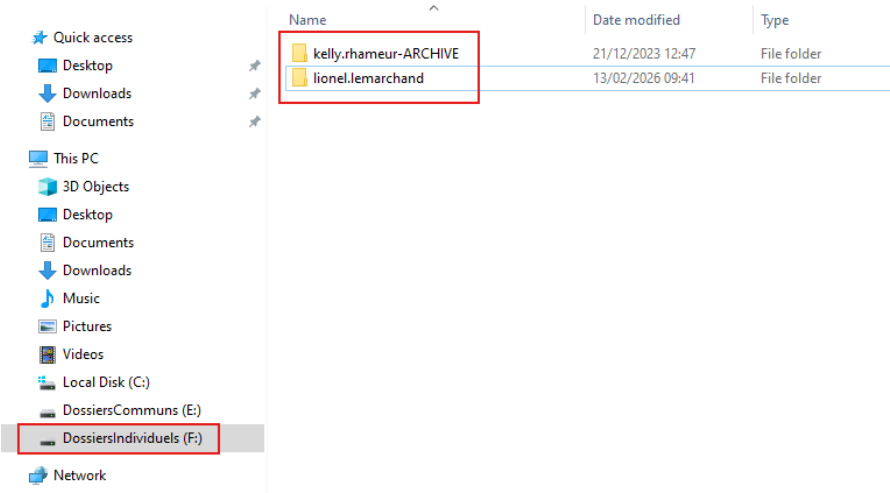
Q.1.1.3

Copie d'écran de l'ancien groupe dans lequel était Kelly Rhameur.



Q.1.1.4

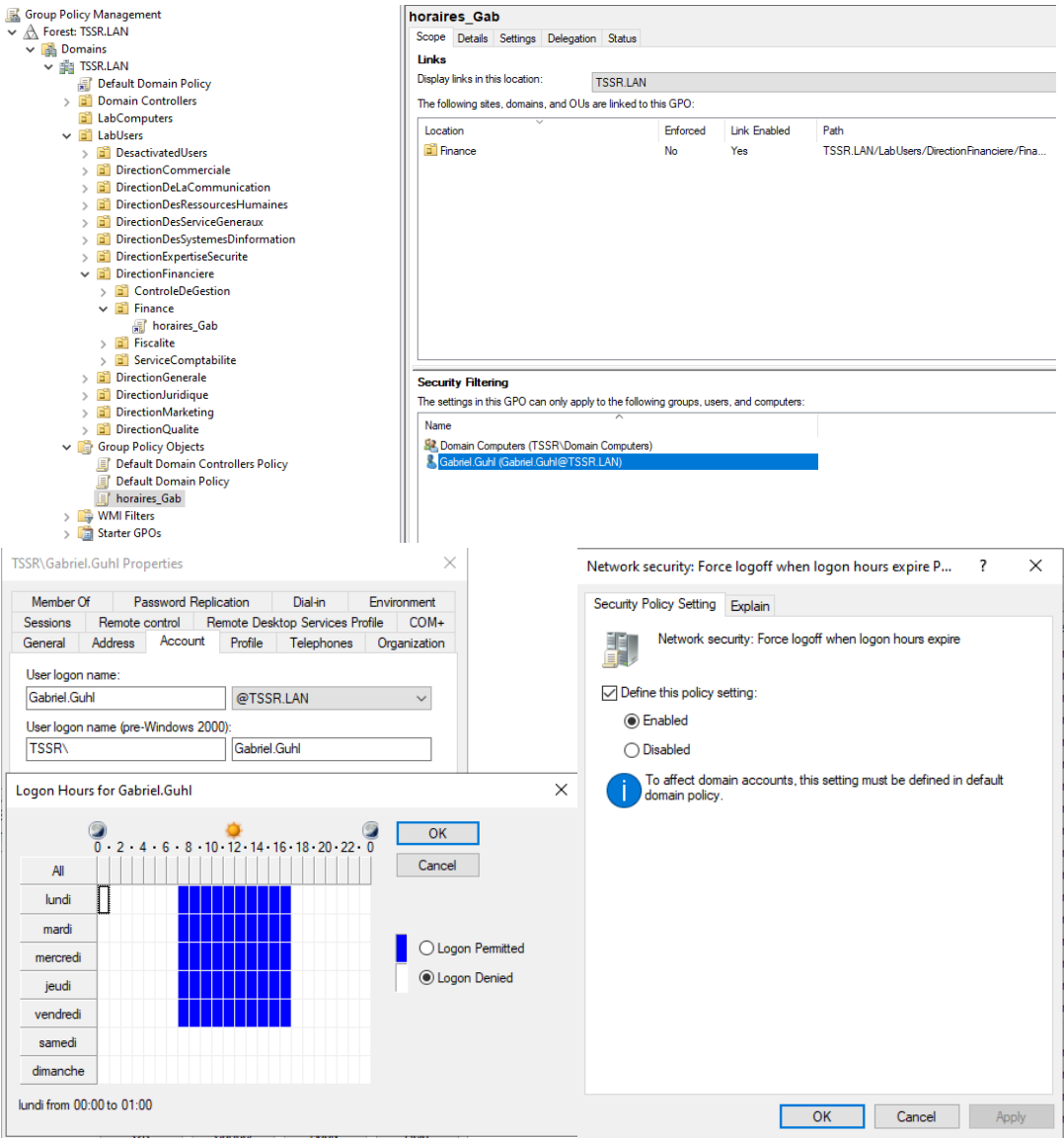
Copie d'écran des dossiers individuels demandés.



Partie 2

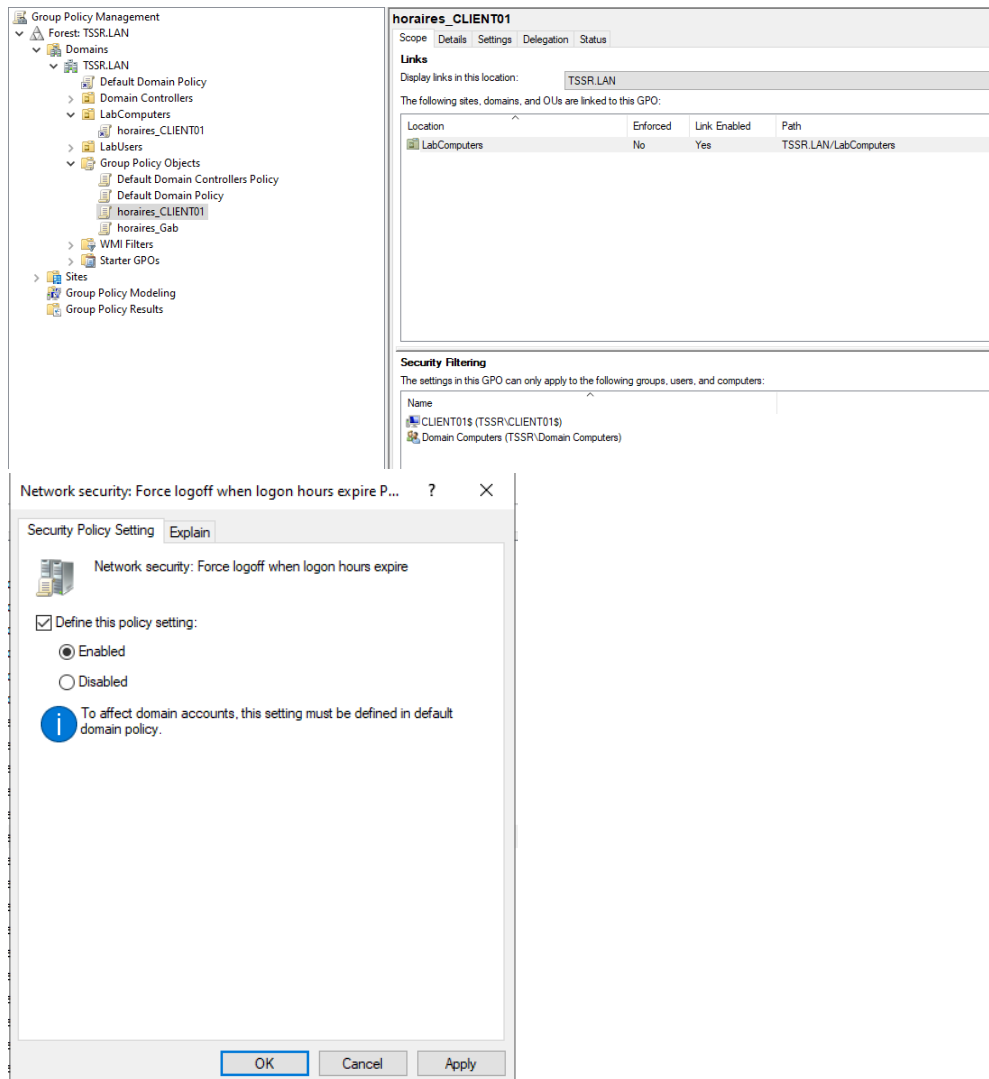
Q.1.2.1

Copies d'écran montrant le paramétrage de la restriction de connexion horaire.



Q.1.2.2

Copie d'écran montrant le paramétrage de la restriction de la connexion machine.



Q.1.2.3

Copies d'écran de la stratégie de mot de passe.

Password must meet complexity requirements Properties

Security Policy Setting Explain

Password must meet complexity requirements

Define this policy setting:
☒ Enabled
☐ Disabled

Minimum password length Properties

Security Policy Setting Explain

Minimum password length

Define this policy setting
Password must be at least:
14 characters

Enforce password history Properties

Security Policy Setting Explain

Enforce password history

Define this policy setting
Keep password history for:
5 passwords remembered

Group Policy Management

Forest: TSSR.LAN

Domains

TSSR.LAN

Default Domain Policy

Domain Controllers

LabComputers

LabUsers

MDP_Users

DesactivatedUsers

DirectionCommerciale

DirectionDeLaCommunication

DirectionDesRessourcesHumaines

DirectionDesServiceGeneraux

DirectionDesSystemesDinformation

DirectionExpertiseSecurite

DirectionFinanciere

DirectionGenerale

DirectionJuridique

DirectionMarketing

DirectionQualite

Group Policy Objects

Default Domain Controllers Policy

Default Domain Policy

horaires_CLIENT01

horaires_Gab

MDP_Users

WMI Filters

Starter GPOs

MDP_Users

Scope Details Settings Delegation Status

Links

Display links in this location: TSSR.LAN

The following sites, domains, and OUs are linked to this GPO:

Location	Enforced	Link Enabled	Path
LabUsers	No	Yes	TSSR.LAN/LabUsers

Security Filtering

The settings in this GPO can only apply to the following groups, users, and computers:

Name
Domain Computers (TSSR\Domain Computers)

Partie 3

Q.1.3.1

Copies d'écran de la GPO de montage de lecteurs.

TSSR.LAN

Default Domain Policy

Domain Controllers

LabComputers

Drive_Mount

horaires_CLIENT01

LabUsers

MDP_Users

DesactivatedUsers

DirectionCommerciale

DirectionDeLaCommunication

DirectionDesRessourcesHumaines

DirectionDesServiceGeneraux

DirectionDesSystemesDinformation

DirectionExpertiseSecurite

DirectionFinanciere

DirectionGenerale

DirectionJuridique

DirectionMarketing

DirectionQualite

Group Policy Objects

Default Domain Controllers Policy

Default Domain Policy

Drive_Mount

horaires_CLIENT01

horaires_Gab

MDP_Users

WMI Filters

Starter GPOs

Links

Display links in this location: TSSR.LAN

The following sites, domains, and OUs are linked to this GPO:

Location	Enforced	Link Enabled	Path
LabComputers	No	Yes	TSSR.LAN/LabComputers

Security Filtering

The settings in this GPO can only apply to the following groups, users, and computers:

Name
Domain Computers (TSSR\Domain Computers)

E: Properties

General

Common

Action: Update

Location: \\SRVWIN01\DossiersCommuns

Reconnect: ☐ Label as: DossiersCommuns

Drive Letter

☐ Use first available, starting at:

☒ Use: E

Connect as (optional)

User name:

Password:

Confirm password:

Hide/Show this drive

☒ No change

☐ Hide this drive

☐ Show this drive

Hide/Show all drives

☒ No change

☐ Hide all drives

☐ Show all drives

OK

Cancel

Apply

Help

F: Properties

General

Common

Action: Update

Location: \\SRVWIN01\DossiersIndividuel

Reconnect: ☐ Label as: DossiersIndividuel

Drive Letter

☐ Use first available, starting at:

☒ Use: F

Connect as (optional)

User name:

Password:

Confirm password:

Hide/Show this drive

☒ No change

☐ Hide this drive

☐ Show this drive

Hide/Show all drives

☒ No change

☐ Hide all drives

☐ Show all drives

• OK

Cancel

Apply

Help

Exercice 2

Partie 1

Q.2.1.1

Copie d'écran de la création de compte.

```
root@SRVLX01:~# useradd -m -s /bin/bash romain
root@SRVLX01:~# passwd romain
Nouveau mot de passe :
Retapez le nouveau mot de passe :
passwd: password updated successfully
root@SRVLX01:~#
```

Q.2.1.2

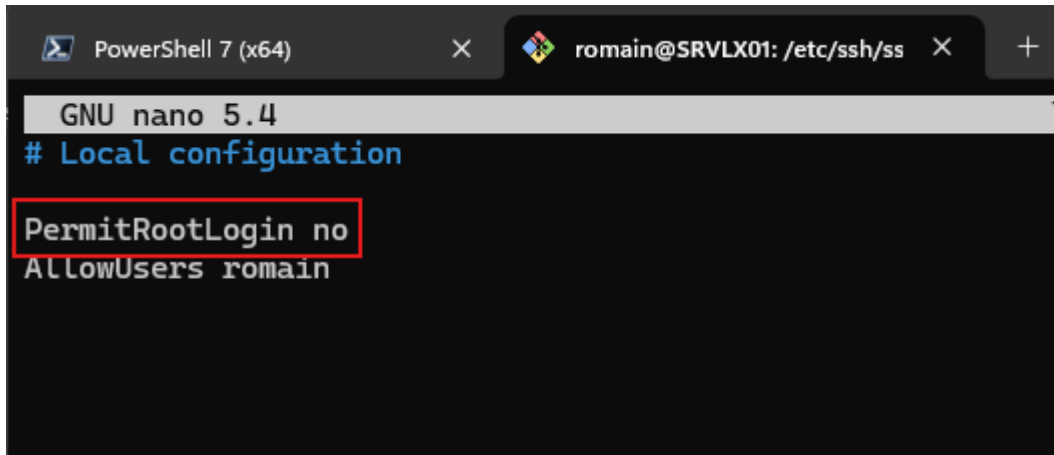
Copie d'écran du paramétrage du compte.

```
root@SRVLX01:~# usermod -aG sudo romain
root@SRVLX01:~#
```

Partie 2

Q.2.2.1

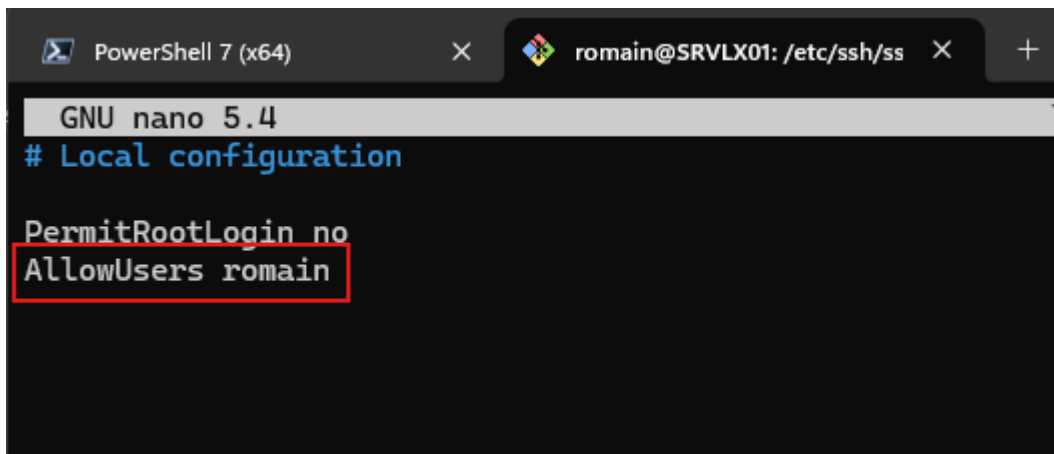
Copie d'écran du paramétrage de l'accès distant.



```
PowerShell 7 (x64)  romain@SRVLX01: /etc/ssh/sshd_config
GNU nano 5.4
# Local configuration
PermitRootLogin no
AllowUsers romain
```

Q.2.2.2

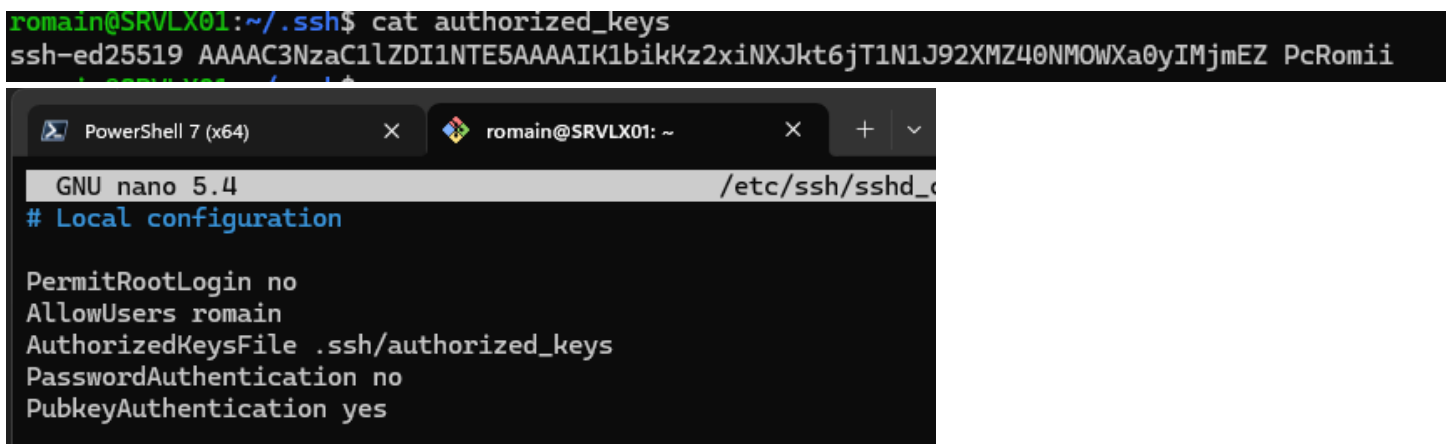
Copie d'écran du paramétrage distant avec le compte personnel.



```
PowerShell 7 (x64)  romain@SRVLX01: /etc/ssh/sshd_config
GNU nano 5.4
# Local configuration
PermitRootLogin no
AllowUsers romain
```

Q.2.2.3

Copies d'écran du paramétrage de l'authentification.

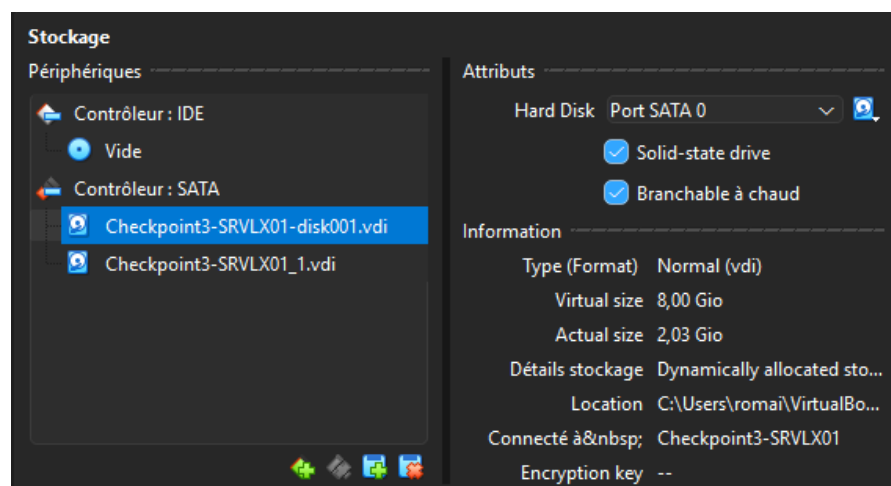


```
romain@SRVLX01:~/ssh$ cat authorized_keys
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIK1bikKz2xiNXJkt6jt1N1J92XMZ40NMOWXa0yIMjmEZ PcRomii
```

```
PowerShell 7 (x64)  romain@SRVLX01: ~
GNU nano 5.4 /etc/ssh/sshd_config
# Local configuration
PermitRootLogin no
AllowUsers romain
AuthorizedKeysFile .ssh/authorized_keys
PasswordAuthentication no
PubkeyAuthentication yes
```


Q.2.3.3

Copies d'écran montrant les différentes étapes pour la réparation du volume RAID.



```
romain@SRVLX01:~$ sudo mdadm --manage /dev/md0 --add /dev/sdb1
mdadm: added /dev/sdb1
romain@SRVLX01:~$ sudo fdisk /dev/sdb
```

```
romain@SRVLX01:~$ lsblk
NAME                MAJ:MIN RM   SIZE RO TYPE MOUNTPOINT
sda                  8:0    0    8G  0 disk
├─sda1               8:1    0    8G  0 part
│   └─md0            9:0    0    8G  0 raid1
│       ├─md0p1      259:0    0 488,3M  0 part /boot
│       ├─md0p2      259:1    0    1K  0 part
│       └─md0p5      259:2    0    7,5G  0 part
│           ├─cp3--vg-root 253:0    0    2,8G  0 lvm  /
│           └─cp3--vg-swap_1 253:1    0    976M  0 lvm  [SWAP]
└─sdb                8:16    0    8G  0 disk
   └─sdb1            8:17    0    8G  0 part
sr0                 11:0    1 1024M  0 rom
```

```
romain@SRVLX01:~$ lsblk
NAME                MAJ:MIN RM   SIZE RO TYPE MOUNTPOINT
sda                  8:0    0    8G  0 disk
├─sda1               8:1    0    8G  0 part
│   └─md0            9:0    0    8G  0 raid1
│       ├─md0p1      259:0    0 488,3M  0 part /boot
│       ├─md0p2      259:1    0    1K  0 part
│       └─md0p5      259:2    0    7,5G  0 part
│           ├─cp3--vg-root 253:0    0    2,8G  0 lvm  /
│           └─cp3--vg-swap_1 253:1    0    976M  0 lvm  [SWAP]
└─sdb                8:16    0    8G  0 disk
   └─sdb1            8:17    0    8G  0 part
sr0                 11:0    1 1024M  0 rom
```

```
romain@SRVLX01:~$ cat /proc/mdstat
Personalities : [raid1] [linear] [multipath] [raid0] [raid6] [raid5] [raid4] [raid10]
md0 : active raid1 sdb1[2] sda1[0]
      8381440 blocks super 1.2 [2/2] [UU]

unused devices: <none>
```

```
romain@SRVLX01:~$ sudo mdadm --detail /dev/md0
/dev/md0:
    Version : 1.2
  Creation Time : Tue Dec 20 10:02:28 2022
    Raid Level : raid1
    Array Size : 8381440 (7.99 GiB 8.58 GB)
  Used Dev Size : 8381440 (7.99 GiB 8.58 GB)
    Raid Devices : 2
  Total Devices : 2
 Persistence : Superblock is persistent

    Update Time : Fri Feb 13 12:12:09 2026
      State : clean
  Active Devices : 2
 Working Devices : 2
 Failed Devices : 0
  Spare Devices : 0

Consistency Policy : resync

    Name : cp3:0
   UUID : 32332561:cf16c858:703517e8:81dd5c10
  Events : 3611

   Number   Major   Minor   RaidDevice State
    -----   -----   -----   -
    0         8         1         0     active sync  /dev/sda1
    2         8        17         1     active sync  /dev/sdb1
```

Q.2.3.4

Copies d'écran montrant les différentes étapes de configuration.

```
romain@SRVLX01:~$ sudo vgs
VG      #PV #LV #SN Attr   VSize VFree
cp3-vg   1   2   0 wz--n- 7,51g <3,79g
romain@SRVLX01:~$ sudo lvcreate -L 2G -n sauvegardes cp3-vg
Logical volume "sauvegardes" created.
romain@SRVLX01:~$ sudo vgs
VG      #PV #LV #SN Attr   VSize VFree
cp3-vg   1   3   0 wz--n- 7,51g <1,79g

romain@SRVLX01:~$ sudo mkfs.ext4 /dev/cp3-vg/sauvegardes
mke2fs 1.46.2 (28-Feb-2021)
Creating filesystem with 524288 4k blocks and 131072 inodes
Filesystem UUID: f7480ed2-1781-41bc-b944-5c63fe072452
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

romain@SRVLX01:~$ sudo mkdir -p /var/lib/bareos/storage
romain@SRVLX01:~$ sudo nano /etc/fstab
```

```
GNU nano 5.4 /etc/fstab
# /etc/fstab: static file system information.
#
# Use 'blkid' to print the universally unique identifier for a
# device; this may be used with UUID= as a more robust way to name devices
# that works even if disks are added and removed. See fstab(5).
#
# systemd generates mount units based on this file, see systemd.mount(5).
# Please run 'systemctl daemon-reload' after making changes here.
#
# <file system> <mount point> <type> <options> <dump> <pass>
/dev/mapper/cp3--vg-root / ext4 errors=remount-ro 0 1
# /boot was on /dev/md0p1 during installation
UUID=9bba6d48-3e4b-42a6-bccc-12836de215ec /boot ext2 defaults 0 2
/dev/mapper/cp3--vg-swap_1 none swap sw 0 0
/dev/sr0 /media/cdrom0 udf,iso9660 user,noauto 0 0
/dev/cp3-vg/sauvegardes /var/lib/bareos/storage ext4 defaults 0 2

romain@SRVLX01:~$ mount -a
mount: /var/lib/bareos/storage: seul le superutilisateur peut utiliser mount.
romain@SRVLX01:~$ sudo mount -a
romain@SRVLX01:~$ lsblk
NAME                                MAJ:MIN RM   SIZE RO TYPE MOUNTPOINT
sda                                  8:0    0    8G  0 disk
├─sda1                               8:1    0    8G  0 part
│   └─md0                            9:0    0    8G  0 raid1
│       ├─md0p1                     259:0  0 488,3M  0 part /boot
│       ├─md0p2                     259:1  0    1K  0 part
│       └─md0p5                     259:2  0    7,5G  0 part
│           ├─cp3--vg-root           253:0  0    2,8G  0 lvm /
│           ├─cp3--vg-swap_1         253:1  0   976M  0 lvm [SWAP]
│           └─cp3--vg-sauvegardes    253:2  0    2G  0 lvm /var/lib/bareos/storage
sdb                                  8:16    0    8G  0 disk
├─sdb1                               8:17    0    8G  0 part
│   └─md0                            9:0    0    8G  0 raid1
│       ├─md0p1                     259:0  0 488,3M  0 part /boot
│       ├─md0p2                     259:1  0    1K  0 part
│       └─md0p5                     259:2  0    7,5G  0 part
│           ├─cp3--vg-root           253:0  0    2,8G  0 lvm /
│           ├─cp3--vg-swap_1         253:1  0   976M  0 lvm [SWAP]
│           └─cp3--vg-sauvegardes    253:2  0    2G  0 lvm /var/lib/bareos/storage
sr0                                  11:0    1 1024M  0 rom
```

Q.2.3.5

Copie d'écran montrant l'espace disponible.

```
romain@SRVLX01:~$ df -h /var/lib/bareos/storage/
Sys. de fichiers      Taille Utilisé Dispo Uti% Monté sur
/dev/mapper/cp3--vg-sauvegardes 2,0G  24K  1,8G  1% /var/lib/bareos/storage
```

Partie 4

Q.2.4.1

Réponse à la question

Partie 5

Q.2.5.1

Réponse à la question

Q.2.5.2

Réponse à la question

Q.2.5.3

Réponse à la question

Q.2.5.4

Réponse à la question

Partie 6

Q.2.6.1

Réponse à la question